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НАЦИОНАЛЬНЫЙ ИССЛЕДОВАТЕЛЬСКИЙ
ТОМСКИЙ ГОСУДАРСТВЕННЫЙ УНИВЕРСИТЕТ (НИ ТГУ)
Институт прикладной математики и информатики

ОТЧЕТ ПО ЛАБОРАТОРНОЙ РАБОТЕ
по дисциплине «Системы управления базами данных»

DDL DML

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Специалитет, 10.05.01,
Направленность «Компьютерная Безопасность»

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подпись

«10» января 2023 г.

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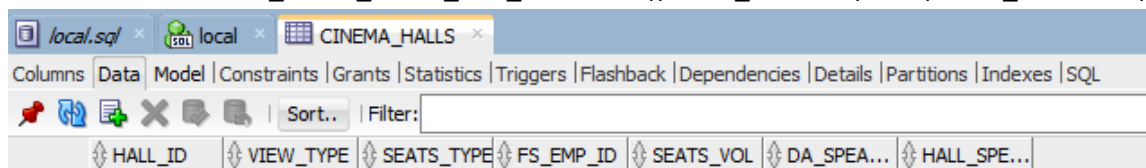
_____ В.О. Бриц

подпись

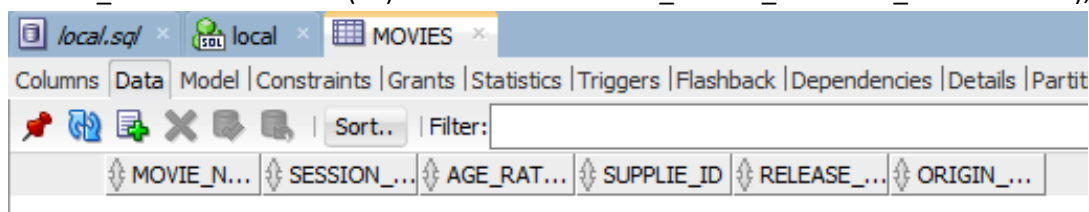
«10» января 2023 г.

1. Создать 3 таблицы со всеми типами ограничений.

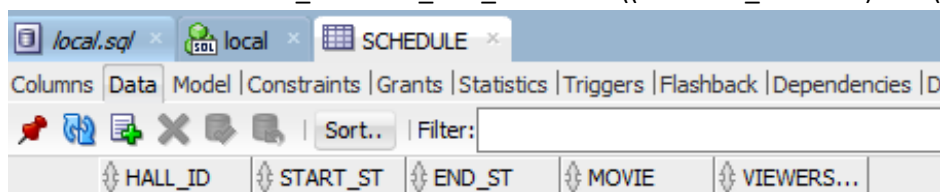
```
CREATE TABLE HR.CINEMA_HALLS
(HALL_ID NUMBER(1) CONSTRAINT CINEMA_HALLS_HALL_ID_NN NOT NULL,
VIEW_TYPE VARCHAR2(2) CONSTRAINT CINEMA_HALLS_VIEW_TYPE_NN NOT NULL,
SEATS_TYPE VARCHAR2(7) CONSTRAINT CINEMA_HALLS_SEATS_TYPE_NN NOT NULL,
FS_EMP_ID NUMBER(3) CONSTRAINT CINEMA_HALLS_FS_EMP_ID_NN NOT NULL UNIQUE,
SEATS_VOL NUMBER(3) CONSTRAINT CINEMA_HALLS_SEATS_VOL_NN NOT NULL,
DA_SPEAKER_VOL NUMBER(2),
HALL_SPEC_NAME VARCHAR2(10),
CONSTRAINT CINEMA_HALLS_HALL_ID_PK PRIMARY KEY(HALL_ID),
CONSTRAINT CINEMA_HALLS_SEATS_VOL_CC CHECK ((SEATS_VOL >= 0) and (SEATS_VOL < 100)));
```



```
CREATE TABLE HR.MOVIES
(MOVIE_NAME VARCHAR2(20) CONSTRAINT MOVIES_MOVIE_NAME_PK PRIMARY KEY,
SESSION_TIME TIMESTAMP CONSTRAINT MOVIES_SESSION_TIME_NN NOT NULL,
AGE_RATING VARCHAR2(3) DEFAULT '0+',
SUPPLIE_ID NUMBER(6) CONSTRAINT MOVIES_SUPPLIER_NN NOT NULL,
RELEASE_YEAR NUMBER(4),
ORIGIN_COUTRY VARCHAR2(10) CONSTRAINT MOVIES_ORIGIN_COUNTRY_NN NOT NULL);
```



```
CREATE TABLE HR.SCHEDULE
(HALL_ID NUMBER(1) CONSTRAINT SCHEDULE_HALL_ID_NN NOT NULL,
START_ST TIMESTAMP CONSTRAINT SCHEDULE_START_ST_NN NOT NULL,
END_ST TIMESTAMP CONSTRAINT SCHEDULE_END_ST_NN NOT NULL,
MOVIE VARCHAR2(20) CONSTRAINT SCHEDULE_MOVIE_NN NOT NULL,
VIEWERS_VOL NUMBER(3) CONSTRAINT SCHEDULE_VIEWERS_VO_NN NOT NULL,
CONSTRAINT SCHEDULE_SESSION_PK PRIMARY KEY(HALL_ID, START_ST),
CONSTRAINT SCHEDULE_HALL_ID_FK FOREIGN KEY(HALL_ID) REFERENCES HR.CINEMA_HALLS(HALL_ID),
CONSTRAINT SCHEDULE_MOVIE_FK FOREIGN KEY(MOVIE) REFERENCES HR.MOVIES(MOVIE_NAME),
CONSTRAINT SCHEDULE_VIEWERS_VOL_CC CHECK ((VIEWERS_VOL >= 0) and (VIEWERS_VOL < 100)));
```



2. Изменить структуру таблицы.

ALTER TABLE HR.SCHEDULE ADD EXP_RATING NUMBER(3);

COLUMN_NAME	DATA_TYPE	NULLABLE	DATA_DEFAULT	COLUMN_ID	COMMENTS
1 HALL_ID	NUMBER (1,0)	No	(null)	1 (null)	
2 START_ST	TIMESTAMP (6)	No	(null)	2 (null)	
3 END_ST	TIMESTAMP (6)	No	(null)	3 (null)	
4 MOVIE	VARCHAR2 (20 BYTE)	No	(null)	4 (null)	
5 VIEWERS_VOL	NUMBER (3,0)	No	(null)	5 (null)	
6 EXP_RATING	NUMBER (3,0)	Yes	(null)	6 (null)	

ALTER TABLE HR.SCHEDULE MODIFY EXP_RATING VARCHAR2(6);

COLUMN_NAME	DATA_TYPE	NULLABLE	DATA_DEFAULT	COLUMN_ID	COMMENTS
1 HALL_ID	NUMBER (1,0)	No	(null)	1 (null)	
2 START_ST	TIMESTAMP (6)	No	(null)	2 (null)	
3 END_ST	TIMESTAMP (6)	No	(null)	3 (null)	
4 MOVIE	VARCHAR2 (20 BYTE)	No	(null)	4 (null)	
5 VIEWERS_VOL	NUMBER (3,0)	No	(null)	5 (null)	
6 EXP_RATING	VARCHAR2 (6 BYTE)	Yes	(null)	6 (null)	

3. Удалить столбцы и таблицу

ALTER TABLE HR.SCHEDULE DROP COLUMN EXP_RATING;

COLUMN_NAME	DATA_TYPE	NULLABLE	DATA_DEFAULT	COLUMN_ID	COMMENTS
1 HALL_ID	NUMBER (1,0)	No	(null)	1 (null)	
2 START_ST	TIMESTAMP (6)	No	(null)	2 (null)	
3 END_ST	TIMESTAMP (6)	No	(null)	3 (null)	
4 MOVIE	VARCHAR2 (20 BYTE)	No	(null)	4 (null)	
5 VIEWERS_VOL	NUMBER (3,0)	No	(null)	5 (null)	

DROP TABLE HR.SCHEDULE;

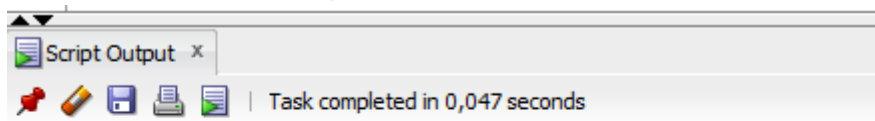


Table HR.SCHEDULE dropped.

4. Вставить данные в созданные таблицы.

INSERT INTO HR.CINEMA_HALLS (HALL_ID, VIEW_TYPE, SEATS_TYPE, FS_EMP_ID, SEATS_VOL)

VALUES (8, '3D', 'DEFAULT', 109, 80);

HALL_ID	VIEW_TYPE	SEATS_TYPE	FS_EMP_ID	SEATS_VOL	DA_SPEAKER_VOL	HALL_SPEC_NAME
1	8 3D	DEFAULT	109	80	(null)	(null)

INSERT INTO HR.MOVIES (MOVIE_NAME, SESSION_TIME, SUPPLIE_ID, ORIGIN_COUTRY)
VALUES
('Knifes out 2', TO_TIMESTAMP('2023/01/19 9:00:00', 'YYYY/MM/DD HH:MI:SS'), 108, 'UK');

MOVIE_NAME	SESSION_TIME	AGE_RATING	SUPPLIE_ID	RELEASE_YEAR	ORIGIN_COUTRY
1 Knifes out 2	19.01.23 09:00:00,000000000	0+	108	(null)	UK

INSERT INTO HR.SCHEDULE VALUES
(8, TO_TIMESTAMP('2023/01/19 9:00:00', 'YYYY/MM/DD HH:MI:SS'),
TO_TIMESTAMP('2023/01/19 11:15:00', 'YYYY/MM/DD HH:MI:SS'), 'Knifes out 2', 0);

HALL_ID	START_ST	END_ST	MOVIE	VIEWERS_VOL
1	8 19.01.23 09:00:00,000000000	19.01.23 11:15:00,000000000	Knifes out 2	0

5. Добавить данные используя условия.

UPDATE HR.Schedule SET viewers_vol = 13 WHERE movie = 'Knifes out 2';

HALL_ID	START_ST	END_ST	MOVIE	VIEWERS_VOL
1	8 19.01.23 09:00:00,000000000	19.01.23 11:15:00,000000000	Knifes out 2	13
2	8 19.01.23 11:30:00,000000000	19.01.23 01:00:00,000000000	Fast and furious X	0
3	8 19.01.23 01:15:00,000000000	19.01.23 02:20:00,000000000	Bimbo	0

6. Удалить данные используя условия.

DELETE HR.Schedule WHERE movie = 'Knifes out 2';

HALL_ID	START_ST	END_ST	MOVIE	VIEWERS_VOL
1	8 19.01.23 11:30:00,000000000	19.01.23 01:00:00,000000000	Fast and furious X	0
2	8 19.01.23 01:15:00,000000000	19.01.23 02:20:00,000000000	Bimbo	0

7. Ошибки на первичный ключ.

INSERT INTO HR.SCHEDULE VALUES
(7, TO_TIMESTAMP('2023/01/19 9:00:00', 'YYYY/MM/DD HH:MI:SS'),

```
TO_TIMESTAMP('2023/01/19 11:15:00', 'YYYY/MM/DD HH:MI:SS'), 'Knives out 2', 0);
```

```
Error starting at line : 1 in command -
```

```
INSERT INTO HR.SCHEDULE VALUES
```

```
(7, TO_TIMESTAMP('2023/01/19 9:00:00', 'YYYY/MM/DD HH:MI:SS'),
```

```
TO_TIMESTAMP('2023/01/19 11:15:00', 'YYYY/MM/DD HH:MI:SS'), 'Knives out 2', 0)
```

```
Error report -
```

```
ORA-02291: нарушено ограничение целостности (HR.SCHEDULE_HALL_ID_FK) - исходный ключ не найден
```

```
INSERT INTO HR.SCHEDULE(START_ST, END_ST, MOVIE, VIEWERS_VOL) VALUES
```

```
(TO_TIMESTAMP('2023/01/19 9:00:00', 'YYYY/MM/DD HH:MI:SS'),
```

```
TO_TIMESTAMP('2023/01/19 11:15:00', 'YYYY/MM/DD HH:MI:SS'), 'Knives out 2', 0);
```

```
Error starting at line : 1 in command -
```

```
INSERT INTO HR.SCHEDULE(START_ST, END_ST, MOVIE, VIEWERS_VOL) VALUES
```

```
(TO_TIMESTAMP('2023/01/19 9:00:00', 'YYYY/MM/DD HH:MI:SS'),
```

```
TO_TIMESTAMP('2023/01/19 11:15:00', 'YYYY/MM/DD HH:MI:SS'), 'Knives out 2', 0)
```

```
Error report -
```

```
ORA-01400: невозможно вставить NULL в ("HR"."SCHEDULE"."HALL_ID")
```

8. Ошибки на внешний ключ.

```
INSERT INTO HR.SCHEDULE VALUES
```

```
(8, TO_TIMESTAMP('2023/01/19 9:00:00', 'YYYY/MM/DD HH:MI:SS'),
```

```
TO_TIMESTAMP('2023/01/19 11:15:00', 'YYYY/MM/DD HH:MI:SS'), 'Knives out', 0);
```

```
Error starting at line : 1 in command -
```

```
INSERT INTO HR.SCHEDULE VALUES
```

```
(8, TO_TIMESTAMP('2023/01/19 9:00:00', 'YYYY/MM/DD HH:MI:SS'),
```

```
TO_TIMESTAMP('2023/01/19 11:15:00', 'YYYY/MM/DD HH:MI:SS'), 'Knives out', 0)
```

```
Error report -
```

```
ORA-02291: нарушено ограничение целостности (HR.SCHEDULE_MOVIE_FK) - исходный ключ не найден
```

```
INSERT INTO HR.SCHEDULE(HALL_ID, START_ST, END_ST, VIEWERS_VOL) VALUES
```

```
(8, TO_TIMESTAMP('2023/01/19 9:00:00', 'YYYY/MM/DD HH:MI:SS'),
```

```
TO_TIMESTAMP('2023/01/19 11:15:00', 'YYYY/MM/DD HH:MI:SS'), 0);
```

```
Error starting at line : 1 in command -
```

```
INSERT INTO HR.SCHEDULE(HALL_ID, START_ST, END_ST, VIEWERS_VOL) VALUES
```

```
(8, TO_TIMESTAMP('2023/01/19 9:00:00', 'YYYY/MM/DD HH:MI:SS'),
```

```
TO_TIMESTAMP('2023/01/19 11:15:00', 'YYYY/MM/DD HH:MI:SS'), 0)
```

```
Error report -
```

```
ORA-01400: невозможно вставить NULL в ("HR"."SCHEDULE"."MOVIE")
```

9. Ошибка с ограничительным интервалом.

```
INSERT INTO HR.SCHEDULE VALUES
```

```
(8, TO_TIMESTAMP('2023/01/19 9:00:00', 'YYYY/MM/DD HH:MI:SS'),
```

```
TO_TIMESTAMP('2023/01/19 11:15:00', 'YYYY/MM/DD HH:MI:SS'), 'Knives out 2', 101);
```

```
Error starting at line : 1 in command -
INSERT INTO HR.SCHEDULE VALUES
(8, TO_TIMESTAMP('2023/01/19 9:00:00', 'YYYY/MM/DD HH:MI:SS'),
TO_TIMESTAMP('2023/01/19 11:15:00', 'YYYY/MM/DD HH:MI:SS'), 'Knives out 2', 101)
Error report -
ORA-02290: нарушено ограничение целостности CHECK(HR.SCHEDULE_VIEWERS_VOL_CC)
```

10. Ошибка уникальности.

```
INSERT INTO HR.CINEMA_HALLS (HALL_ID, VIEW_TYPE, SEATS_TYPE, FS_EMP_ID, SEATS_VOL)
VALUES (8, '2D', 'DEFAULT', 109, 60);
```

```
Error starting at line : 1 in command -
INSERT INTO HR.CINEMA_HALLS (HALL_ID, VIEW_TYPE, SEATS_TYPE, FS_EMP_ID, SEATS_VOL)
VALUES (8, '2D', 'DEFAULT', 109, 60)
Error report -
ORA-00001: нарушено ограничение уникальности (HR.CINEMA_HALLS_HALL_ID_PK)
```